Lecture 9: Likelihood Ratio Statistics and p-Values

IEOR E4709: Statistic Analysis and Time Series
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1 Likelihood Ratio Tests

The Neyman-Pearson Lemma suggests likelihood ratio tests of the following form. Consider a k-dimensional parameter space $\Theta_0 \subset \Theta$. We can denote by $\theta = (\theta_1, \ldots, \theta_k)$ an element of the parameter space Θ. Define the test statistics

$$\lambda(x) := \frac{\sup_{\theta \in \Theta_0} L(\theta; x)}{\sup_{\theta \in \Theta} L(\theta; x)}$$

For a realization x, determine its best chance of occurrence under $H_0 : \theta \in \Theta_0$, and also its best chance overall. The ratio of these two chances can never exceed unity. If small, it provides evidence for rejection of the null hypothesis. A likelihood ratio test for testing $H_0 : \theta \in \Theta_0$ versus $H_1 : \theta \notin \Theta_0$ is a test whose rejection region (sometimes also referred to as the critical region) is of the form

$$C_1 = \{x : \lambda(x) \leq k\},$$

where $0 \leq k \leq 1$. The test will be at the significance level α if k can be chosen to satisfy

$$\sup \{\mathbb{P}_{\theta \in \Theta_0}(\lambda(X) \leq k)\} = \alpha,$$

i.e. the probability of rejecting the null hypothesis H_0 when it is true is equal to α. If the null hypothesis is simple, that is $\Theta_0 = \{\theta_0\}$, we obtain the simpler form $\mathbb{P}_{\theta_0}(\lambda(X) \leq k) = \alpha$. To determine k, we need to characterize the cumulative distribution function of the r.v. $\lambda(X)$, when the random sample $X = (x_1, x_2, \ldots, x_n)$ has joint pdf $L(\theta_0; x)$.

Example: The null hypothesis is $H_0 : \theta = \theta_0$ for the mean of a normal distribution with unknown variance σ^2. Hence, $\Theta_0 = \left\{(\theta, \sigma^2) : \theta = \theta_0, \sigma^2 \in \mathbb{R}^+\right\}$, and $\Theta = \left\{(\theta, \sigma^2) : \theta \in \mathbb{R}, \sigma^2 \in \mathbb{R}^+\right\}$. The test statistic is of the form

$$\lambda(x) := \frac{\sup_{\sigma^2} L(\theta_0; x)}{\sup_{(\theta, \sigma^2)} L(\theta; x)}$$

Recall that the likelihood of a Gaussian associated with the sample $x = (x_1, \ldots, x_n)$ is

$$L(\theta_0, \sigma^2; x) = (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^n (x_i - \theta_0)^2},$$

and the log-likelihood associated with the sample $x = (x_1, \ldots, x_n)$ is

$$LL(\theta_0, \sigma^2; x) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \theta_0)^2$$

Let us try to find the maximum of LL over σ^2. Taking the derivative with respect with σ^2 and setting it to zero, we find that

$$\frac{\partial LL}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i - \theta_0)^2 = 0$$

is satisfied if $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \theta_0)^2$. Plugging this optimum into the likelihood function, we obtain

$$\sup_{\sigma^2}\left\{L(\theta_0, \sigma^2; x)\right\} = \left(\frac{2\pi}{n}\sum_{i=1}^{n}(x_i - \theta_0)^2\right)^{-n/2} e^{-n/2}$$

For the denominator of the test statistics, we already know that the maximum likelihood estimator of θ is $\bar{x} := \frac{1}{n}\sum_{i=1}^{n} x_i$. Hence,

$$\sup_{\theta, \sigma^2}\left\{L(\theta, \sigma^2; x)\right\} = \left(\frac{2\pi}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2\right)^{-n/2} e^{-n/2}.$$

Then we can rewrite

$$\lambda(x) = \left(\frac{\sum_{i=1}^{n}(x_i - \theta_0)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)^{-n/2}.$$

Notice that

$$\begin{aligned}\sum_{i=1}^{n}(x_i - \theta_0)^2 &= \sum_{i=1}^{n}((x_i - \bar{x}) + (\bar{x} - \theta_0))^2 = \sum_{i=1}^{n}(x_i - \bar{x})^2 + \sum_{i=1}^{n}(\bar{x} - \theta_0)^2 + 2\sum_{i=1}^{n}(x_i - \bar{x})(\bar{x} - \theta_0) \\ &= \sum_{i=1}^{n}(x_i - \bar{x})^2 + n(\bar{x} - \theta_0)^2 + 2(\bar{x} - \theta_0)\sum_{i=1}^{n}(x_i - \bar{x}) = \sum_{i=1}^{n}(x_i - \bar{x})^2 + n(\bar{x} - \theta_0)^2\end{aligned}$$

so that $\lambda(x) = \left(1 + \frac{n(\bar{x}-\theta_0)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)^{-n/2}$. Define

$$\frac{|\bar{x} - \theta_0|}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}}. \tag{1}$$

Recall that the critical region is $C_1 = \{x : \lambda(x) \leq k\}$. So, it follows that H_0 is rejected when the value of the expression in (1) is larger than some constant. We can equivalently write the rejection, also called critical, region as

$$C_1 = \left\{\frac{|\bar{x} - \theta_0|}{s/\sqrt{n}} \geq c\right\},$$

where $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$. Recall from previous lectures and homework problem that

$$\frac{\bar{X} - \theta_0}{S/\sqrt{n}} \sim t_{n-1},$$

where $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$. This is the form of *a two-sided t-test* (as we reject both if the value of the statistic is too low (negative), and if it is too large (positive)). We can then use the quantile of the t_{n-1} distribution and specify the constant c based on the desired significance level α.

2 Likelihood Ratio Statistics and Asymptotic Distribution

Since the function $-2\log\lambda(x)$ is a decreasing function of $\lambda(x)$, it follows that the critical region of the likelihood ratio test can also be written as

$$C_1 = \{x : -2\log\lambda(x) \geq c\}.$$

Why do we focus on the statistics $-2\log\lambda(x)$? In general, it is rather difficult to compute the probability distribution of the test statistics $\lambda(x)$. However, what is remarkable, is that we can asymptotically (when the number of observations is large) estimate the distribution of $-2\log\lambda(x)$. Consider the case when the null hypothesis is simple, i.e. $H_0 : \theta = \theta_0$. Let us first define $\Lambda(x) = -2\log\lambda(x) = -2\Big(LL(\theta_0; x) - LL(\hat{\theta}; x)\Big)$, where $\hat{\theta}$ is the maximum likelihood estimate of the parameter θ. Then the critical region may be written as

$$C_1 = \{\Lambda(x) \geq c\},$$

and $\Lambda(x)$ is usually referred to as the likelihood ratio statistic. Let us now study the asymptotic distribution of the test statistics $\Lambda(X)$. The idea is that values of θ close to the MLE estimate $\hat{\theta}$ should be well supported by the data. Thus, if θ_0 is a possible value of θ, then we can approximate the log-likelihood function at θ_0 using a Taylor expansion around $\hat{\theta}$. Indeed,

$$LL(\theta_0; x) \approx LL(\hat{\theta}; x) + \Big(\theta_0 - \hat{\theta}\Big)\left.\frac{\partial LL(\theta; x)}{\partial\theta}\right|_{\theta=\hat{\theta}} + \frac{1}{2}\Big(\theta_0 - \hat{\theta}\Big)^2\left.\frac{\partial LL(\theta; x)}{\partial^2\theta}\right|_{\theta=\hat{\theta}}.$$

By definition, $\hat{\theta}$ is the MLE and, assuming that the maximum of the likelihood function is an interior point, so that the local maximum also coincides with the global maximum, then $\left.\frac{\partial LL(\theta;x)}{\partial\theta}\right|_{\theta=\hat{\theta}} = 0$. Hence,

$$\begin{aligned}\Lambda &\propto -(\hat{\theta} - \theta_0)^2\left.\frac{\partial LL(\theta; x)}{\partial^2\theta}\right|_{\theta=\hat{\theta}} \\ &= (\hat{\theta} - \theta_0)^2 I(\theta_0)\frac{J(\hat{\theta})}{I(\theta_0)},\end{aligned}$$

where $I(\theta_0)$ is the Fisher information matrix associated with the parameter θ_0 and computed using the n observations at disposal (recall that this is given by $-\sum_{i=1}^{n}\mathbb{E}\left[\frac{\partial LL(\theta;X_i)}{\partial^2\theta}\big|_{\theta=\theta_0}\right]$), while $J(\hat{\theta}) = -\sum_{i=1}^{n}\frac{\partial}{\partial^2\theta}LL(\theta; x_i)\big|_{\theta=\hat{\theta}}$ is the observed Fisher information matrix. We know that

$$(\hat{\theta} - \theta_0)I(\theta_0)^{1/2} \xrightarrow{D} Gaussian(0, 1)$$

under the null hypothesis. Thus $(\hat{\theta} - \theta_0)^2 I(\theta_0) \xrightarrow{D} \chi_1^2$, because the square of a Gaussian random variable follows a χ-squared distribution with one degree of freedom. Moreover, it can be shown

that, under the null hypothesis, $\frac{J(\hat{\theta})}{I(\theta_0)} \xrightarrow{\mathbb{P}} 1$. Using Slutsky's theorem, we can conclude that the likelihood ratio statistic

$$\Lambda \xrightarrow{D} \chi_1^2$$

under the null hypothesis that $\theta = \theta_0$.